

State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts

Insights from JetBrains' annual survey of 23,262 developers worldwide

2024

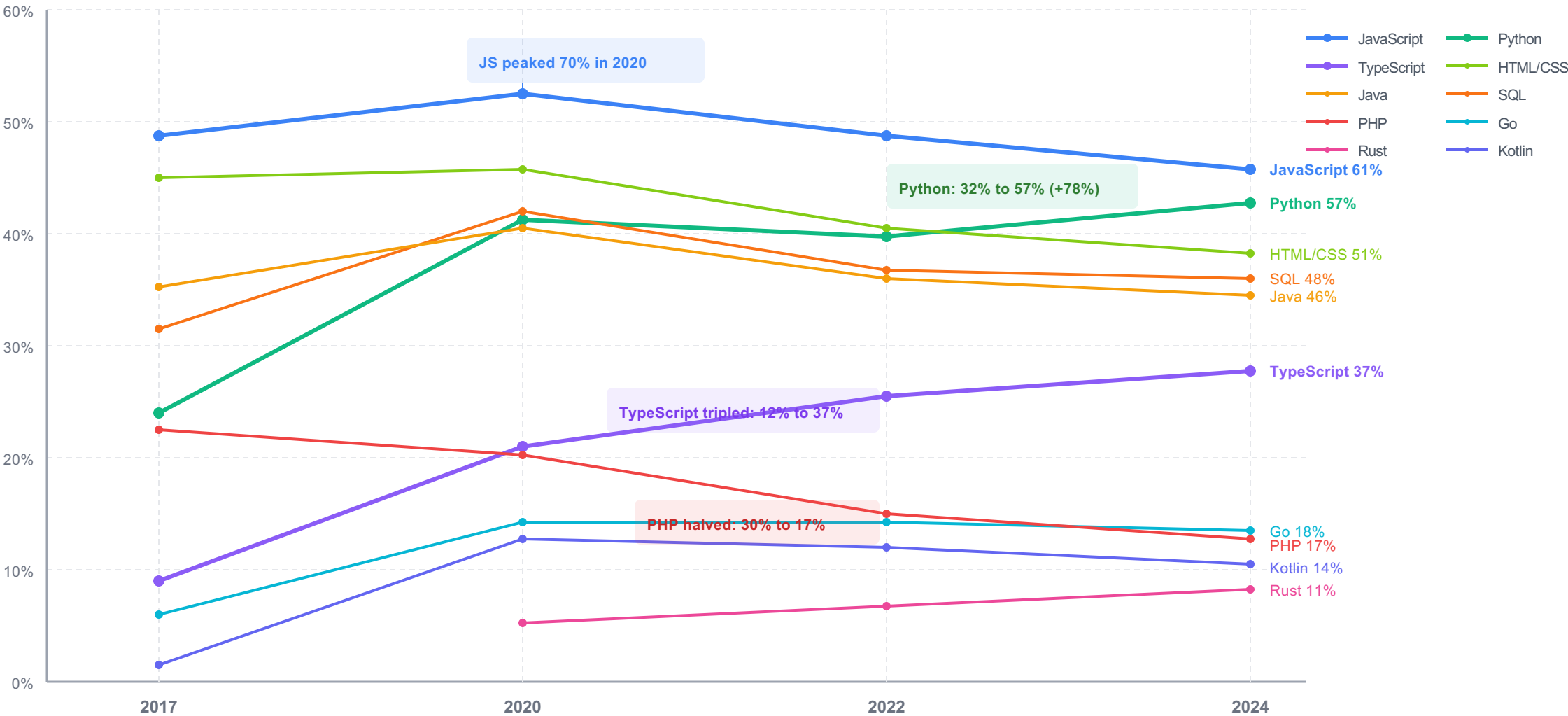
23,262 developers surveyed

20 languages tracked

8 years of trend data

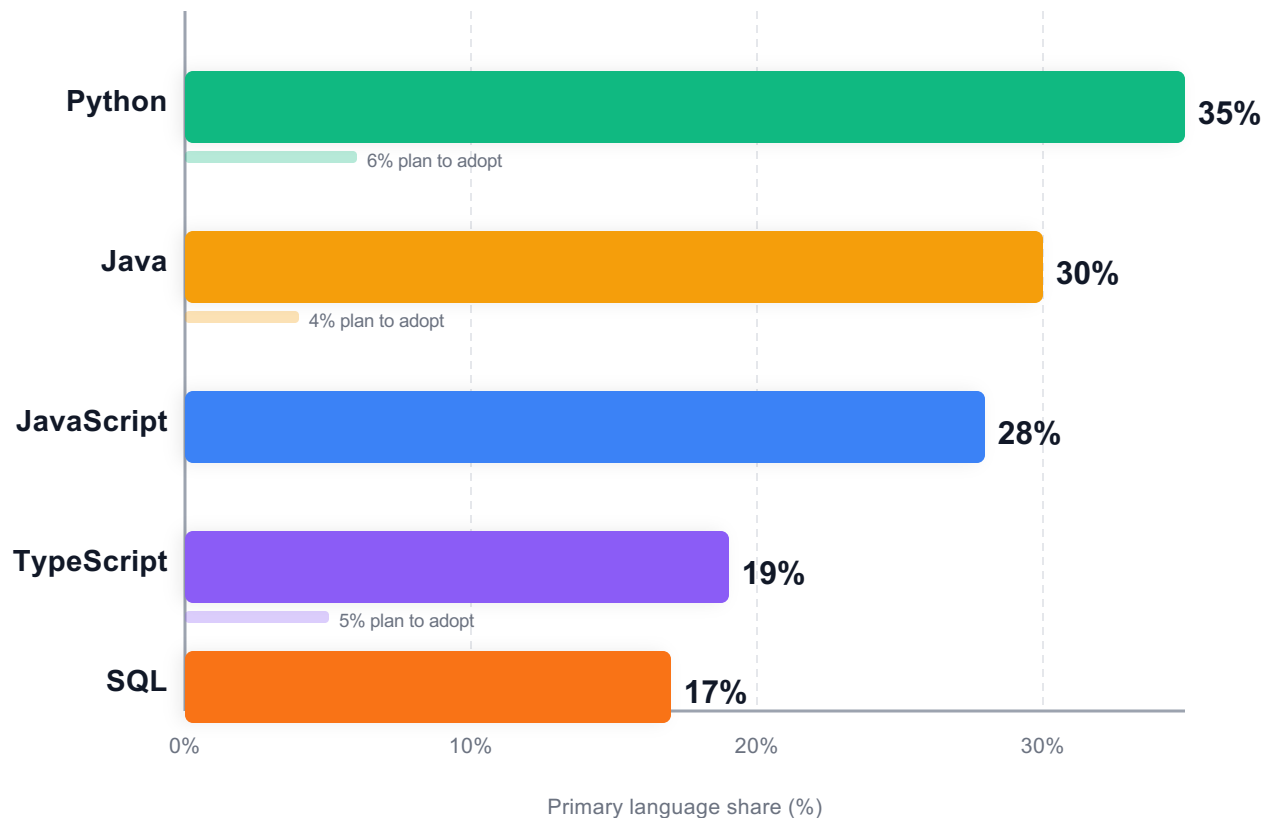
Python Closes the Gap: 8 Years of Language Usage Shifts

JavaScript peaked at 70% in 2020; Python rose from 32% to 57%; TypeScript tripled to 37%; PHP halved to 17%



Python Is Now the #1 Primary Language at 35%

Adoption intent stays below 6% across all languages — developers stick with what they know



Ecosystem Lock-in

Only 6% plan to adopt Python as primary.
Only 5% plan to adopt TypeScript.
Only 4% plan to adopt Java or SQL.

Developers rarely switch primary stacks.

The Comfort Zone Effect

Expertise investment and ecosystem lock-in keep developers loyal to their chosen stacks.

Hiring for niche languages requires active sourcing.

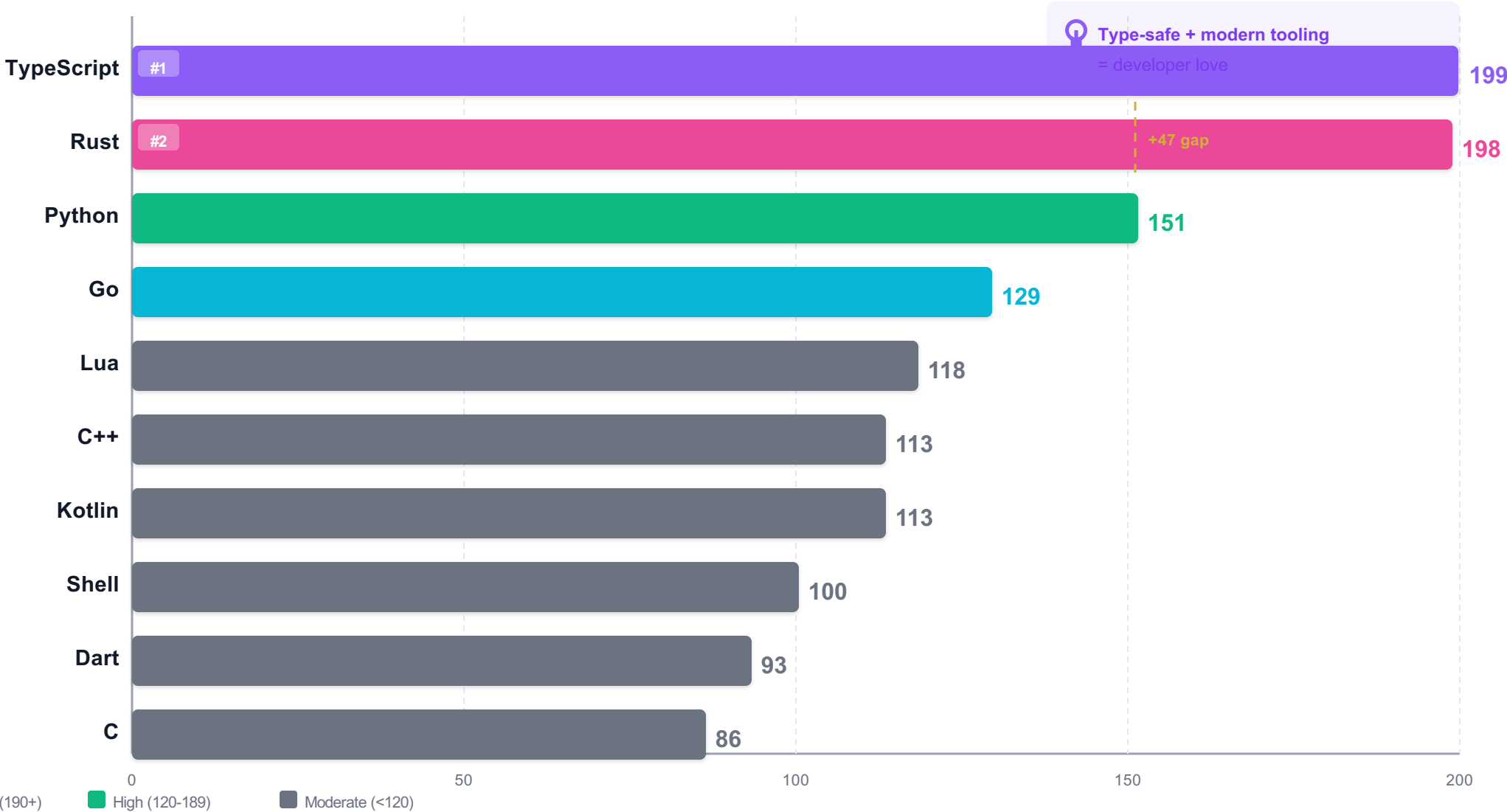
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adoption intent across all languages

Developers stick with what they know

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Type-safe, modern-tooling languages dominate the top spots — satisfaction correlates with developer experience quality



Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Language choice is driven by domain specialization, not general preference — hire for the domain, not the language

APPLICATION DOMAIN	LEADING LANGUAGE	SHARE
 AI / Machine Learning	Python	31%
 Data Processing & Analytics	Python	17%
 User Interface	TypeScript	62%
 APIs & Services	Java	49%
 Application Logic & Workflows	Java	55%
 System Tools & Components	C++	27%
 Integrating with APIs & Services	Python	38%

 Python = AI & Data

AI/ML: 31% | Data Science: 28%
Data Processing: 17% | API Integration: 38%

Undisputed leader in AI and data domains

 TypeScript/JS = UI

UI: TypeScript 62% | JavaScript 58%

Dominates frontend and full-stack

 Java = Backend Services

APIs: 49% | App Logic: 55%

Enterprise backend leader

 C++ = Systems

System Tools: 27% | Hardware: 21%

Performance-critical infrastructure

What This Means for Your Technology Strategy in 2025

Invest in Python for AI/ML

Python is the undisputed leader in AI and data science.

AI/ML: 31% market share
Data Science: 28% market share
API Integration: 38% market share

PRIORITY

TypeScript Is the Future

The future of frontend and full-stack development.

Satisfaction leader: score of 199
Usage tripled: 12% to 37%
UI dominance: 62% market share

PRIORITY

Rust & Go: Growth Plays

The growth plays for infrastructure and systems engineering.

Rust: climbing to 11%, satisfaction 198
Go: stable at 18%, performance focus
Both: top adoption intent languages

GROWTH

PHP & Ruby Hiring Pools Are Shrinking

Plan for succession — these talent pools are contracting.

PHP: dropped from 30% to 17% (2017-2024)
Ruby: fell from 10% to 4% over the same period

WARNING

Satisfaction Drives Retention

Developer satisfaction correlates with type safety and modern tooling — factor this into retention.

Top 2 satisfaction: TypeScript (199), Rust (198)
Both combine type safety with modern tooling

INSIGHT

Core insight: Hire for domain, invest in type-safe tooling, plan for declining language ecosystems